ClawFarm: A Protocol for Mining AI Inference

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Abstract

ClawFarm is a protocol for mining AI inference. Providers register endpoints, users deposit USDC into non-custodial escrow, and each settled call produces a dual-signed proof that determines payment and CLAF emission. Providers select a protocol-fee tier between 0.5 percent and 3.0 percent, in 0.5 percent increments. For settlement amount A and selected rate r, treasury receives A x r and provider pending revenue receives A x (1 - r). CLAF emission follows a ten-year halving schedule, with 70 percent assigned to providers and 30 percent assigned to developers and users, weighted by actual protocol-fee contribution. The protocol does not verify model identity or inspect inference content. It is source-blind by design, allowing any wallet, endpoint, and inference source to enter the same settleable network.

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1. Purpose

ClawFarm is a protocol for mining AI inference. It treats inference as an economic output that can be metered, priced, settled, and rewarded without requiring a platform operator to decide who may supply it or who may consume it.

The central risk of the AGI era is not that models become more capable. It is that model capability, compute access, settlement rails, and reward distribution become locked inside a small number of institutional balance sheets. If the supply side of machine intelligence is captured, every application above it becomes downstream of the same narrow choke point.

ClawFarm is designed to separate the supply side of machine intelligence from any single company, marketplace, or account system. Any wallet, any endpoint, and any source of inference capacity can enter the same settleable network, subject only to the protocol's proof, bond, settlement, and challenge rules.

2. What the protocol is

ClawFarm is not a model lab, a cloud provider, an inference reseller, or a hosted application. It is a settlement protocol for inference calls. The protocol records provider registration, escrowed user funds, dual-signed usage proofs, settlement events, treasury inflows, mining rewards, and burn events.

Applications use a chat-completion compatible interface. Providers register an endpoint and declare model, price, and quality information. A call settles only after the user and provider both sign the same usage proof. The protocol does not inspect the inference itself and does not claim to verify model identity.

This narrowness is deliberate. A protocol that decides which provider is legitimate becomes a platform. A protocol that only settles signed usage can remain identity-blind, source-blind, and forkable.

3. Supply neutrality

Inference capacity is not a single kind of resource. It may come from closed-model API resale, subscription credit pools, self-hosted open-weight models, leased GPU capacity, colocated infrastructure, or future sources that do not exist yet. The protocol does not record which source a provider uses.

This is a structural property, not a slogan. Registration asks for a wallet, a bond, an endpoint, and declared offerings. Settlement asks for a dual-signed proof. Reward accounting reads settled volume and price. None of these operations requires the protocol to ask where capacity came from.

Supply neutrality is the mechanism by which AGI capacity can be opened at scale. The protocol cannot make compute abundant by itself. It can remove the settlement and reward bottleneck that would otherwise force capacity through a small number of intermediaries.

4. Roles

A developer or user deposits USDC into an escrow PDA and consumes inference through an application or SDK. Funds leave escrow only through withdrawal or settlement against a dual-signed proof.

A provider registers an endpoint by posting a 100 USDC bond. The provider selects a protocol-fee tier from 0.5 percent to 3.0 percent, in 0.5 percent increments. The provider receives the settlement remainder after the selected fee and earns CLAF from the Provider Pool according to actual fee contribution, price weight, and quality factors.

A challenger is any party that posts a challenge bond against a suspect settlement. Challenges are permissionless. Enforcement is therefore not operator-driven; it is carried by the economic incentives of participants who can prove a fault.

5. Escrow

User funds are held in a Program Derived Address controlled by the escrow program. No private key controls the escrow account. No administrator can sweep balances. The program releases funds only according to the rules deployed at Genesis.

A user may withdraw available funds at any time, subject to unsettled obligations. A settlement instruction transfers the provider share to the provider wallet and the protocol fee to the treasury PDA. The settlement path is deterministic and does not require human approval.

6. Dual-signed proof

A usage proof binds the request hash, response hash, model identifier, input token count, output token count, agreed price, timestamp, user wallet, and provider wallet. The user signs the proof. The provider signs the same proof. Settlement requires both signatures.

Neither side can unilaterally distort the settlement amount. A provider cannot overstate tokens without the user's signature. A user cannot understate tokens without the provider's signature. The protocol validates signatures and accounting, not the semantic quality of the answer.

This distinction matters. The protocol does not know whether an answer is good, whether a model is the exact version declared, or whether the upstream source is a frontier lab, a subscription account, or a local GPU. It knows that both parties signed the same metered receipt.

7. Settlement

For a settled call, the payment amount A is calculated from the provider's declared price and the metered usage in the signed proof. The provider selects a protocol-fee rate r from the allowed set: 0.5 percent, 1.0 percent, 1.5 percent, 2.0 percent, 2.5 percent, or 3.0 percent. Treasury receives A x r. Provider pending revenue receives A x (1 - r).

The fee tier is provider-selected, not governance-selected. Lower fee tiers reduce the treasury contribution and reduce CLAF reward weight in the same proportion. A provider may choose lower friction for users, but cannot keep the same mining weight while contributing less to the protocol.

The protocol-fee inflow is not an operating budget. It is not allocated to a foundation, contributors, marketing, or maintenance. It is the accounting input that links settlement volume to mining weight and treasury accumulation.

8. Mining and emission

Each settled inference call mines CLAF. Emission is divided between two pools: 70 percent to the providing side and 30 percent to the consuming side. The schedule runs for ten years and halves every two years.

The maximum emitted supply over the schedule is approximately 968.75 million CLAF. The remaining approximately 31.25 million CLAF is never emitted by the protocol. Total supply is fixed at 1,000,000,000 CLAF.

Within each pool, reward weight follows actual protocol-fee contribution rather than nominal settlement volume alone. A call settled at a lower fee tier contributes less mining weight than the same call settled at a higher fee tier. Token distribution therefore follows the amount of USDC fee paid into the protocol.

Provider rewards are additionally weighted by price relative to the network average. A provider that clears below the network average price receives a higher CLAF weight for the same fee contribution. The mechanism subsidizes production of inference when USDC clearing price is below immediate marginal cost.

Developer rewards are earned by settled consumption and inherit the same fee-contribution weighting. This gives the demand side a share of emission and helps bootstrap both sides of the network at the same time.

9. Treasury and burn

The protocol treasury receives the provider-selected fee from every settlement in USDC. The allowed fee tiers range from 0.5 percent to 3.0 percent in 0.5 percent increments. At epoch boundaries, the treasury program evaluates whether accumulated USDC exceeds the configured threshold. If the threshold is met, USDC is swapped for CLAF through Jupiter and the acquired CLAF is burned.

The default epoch length is one hour. The treasury swap threshold is 100 USDC. The slippage cap is 1 percent. The per-swap volume cap is 0.5 percent of relevant pool liquidity.

This mechanism causes circulating supply to contract monotonically in protocol usage, subject to available liquidity and successful swap execution. No human trigger, admin key, or spending committee is involved.

10. Registry and routing

The registry stores provider wallets, endpoints, declared models, pricing, and status. It does not store legal identity, capacity source, or upstream account information. Any wallet that posts the required bond can register.

Routing is performed by clients and applications against registry data. A route may prioritize price, latency, quality history, or a custom policy selected by the developer. The protocol does not operate a central router.

The registry makes supply legible without making it permissioned. Developers can see which models have independent wallet-backed supply and what prices are being cleared, while providers remain peers rather than approved vendors.

11. Bond and challenge

Provider registration requires a 100 USDC bond. The bond creates a cost to spam registration and gives the challenge system an economic anchor. Deregistration includes a cooldown so recent settlements can still be challenged.

Any party may challenge a suspect settlement by posting a small bond. If the challenge is upheld, the provider is slashed and the challenger receives a reward. If the challenge fails, the challenger forfeits the bond. The mechanism discourages both provider abuse and frivolous challenges.

Challenge-driven enforcement is weaker than perfect cryptographic verification of inference, but it is available now and preserves source neutrality. Future cryptographic proof systems can be layered above or forked into a new protocol if they become practical.

12. Immutability

At Genesis, program parameters are initialized and upgrade authority is renounced. After that transaction, no original author, deployer, team, multisig, or governance process can modify the protocol.

Immutability is costly. Bugs cannot be patched in place. Parameters cannot be adjusted to market conditions. The only path to a different rule set is a fork. That cost is accepted because a settlement layer that can be changed by insiders is not neutral infrastructure.

The protocol has no team allocation, no investor allocation, no treasury spending authority, and no governance token rights. Rewards follow settled contribution.

13. Security limits

ClawFarm does not verify model identity. It does not know whether a provider served the exact model declared. It does not know the provider's upstream source. It does not solve all collusion or wash-usage attacks. It imposes costs, signatures, bonds, and challenge incentives.

The protocol depends on Solana for execution, finality, censorship resistance, and token-account semantics. A base-layer failure or halt affects ClawFarm directly.

These limits are not hidden. They define the boundary of the design. ClawFarm is a settlement and mining protocol for inference usage, not a universal truth machine for artificial intelligence.

14. Conclusion

AGI capacity should not be locked behind a small number of accounts, balance sheets, and settlement channels. If intelligence becomes a core economic input, the rails that admit supply and distribute rewards matter as much as the models themselves.

ClawFarm makes inference supply permissionless and settleable. It lets any wallet register capacity, any application consume capacity, and both sides mine CLAF through real settled usage. The protocol does not ask who participates. It asks only that settled calls carry proof.

Compute may centralize. Accounts may centralize. Platforms may centralize. Settlement does not have to.

Appendix A. Genesis parameters

Parameter	Value
Chain	Solana
Settlement asset	USDC
Reward token	CLAF
Total supply	1,000,000,000 CLAF
Maximum scheduled emission	Approx. 968.75M CLAF
Unemitted residual	Approx. 31.25M CLAF
Emission horizon	10 years
Halving interval	2 years
Default epoch length	1 hour
Provider pool	70 percent of epoch emission
Developer pool	30 percent of epoch emission
Provider settlement share	99.5 to 97.0 percent of USDC settlement
Protocol fee	0.5 to 3.0 percent, provider-selected in 0.5 percent increments
Reward weight basis	Proportional to actual USDC protocol fee contributed
Treasury disposition	100 percent buyback-and-burn
Treasury threshold	100 USDC
Swap slippage cap	1 percent
Swap volume cap	0.5 percent of relevant pool liquidity
Provider bond	100 USDC
Challenge mechanism	Permissionless bond and slash
Upgrade authority	Renounced at Genesis
Admin / governance	None

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